# Centroid Excursion of an Apodized Butterfly Field and Its Inversion Pair

Auto BeamScientist Project

**Abstract**

A field and its spatially inverted copy can suppress inversion-odd moments under symmetry conditions, but that identity alone does not determine a finite-interval centroid excursion. We examine the registered comparison for an apodized butterfly field: a coherent copy rotated by 180°, with zero relative phase and amplitude weight 0.8, versus the single source in free space from $z = 0.02$ to 0.06 m. The reference centroid excursions are $0.421 \times 10^{-6}$ m for the coherent pair and $3.708 \times 10^{-6}$ m for the single source, giving a difference of $-3.286 \times 10^{-6}$ m. This nonzero contrast is reproduced across numerical tiers, but the $2.286 \times 10^{-6}$ m margin beyond the registered $1 \times 10^{-6}$ m reduction threshold is smaller than the $3.125 \times 10^{-6}$ m confirmation error floor. The registered error-margin criterion is therefore not passed. A measured weight series and incoherent-mixture comparison show that reduced excursion is not unique to coherent pairing. A conditional parity identity, including overlap-dependent normalization, agrees closely with the saved trajectories under its stated assumptions; it does not establish a general weight law or a coherence-specific mechanism.

## 1 Introduction

The transverse intensity centroid tracks the position of the beam's intensity distribution, rather than the location or value of its brightest pixel. Its trajectory provides a measure of beam motion, but that motion depends on field symmetry and on the domain over which the intensity moments are calculated. Inversion symmetry therefore offers a useful starting point for understanding centroid changes, while leaving open what happens for an unequal-weight pair and a finite observation interval.

We ask whether a coherent, zero-phase inversion pair made from the specified apodized butterfly source reduces centroid excursion relative to the single source over 0.02–0.06 m. The frozen primary condition uses a 180° rotation and amplitude weight $a = 0.8$. The equal-weight pair is a symmetry-limit diagnostic; the zero-weight case checks the single-source identity. The primary observable is the largest displacement of the intensity centroid from its position at the first sampled plane. It is not a measure of peak intensity, collected core energy, or delivered power.

The reference calculation yields a negative coherent-minus-single contrast. Its interpretation depends on distinguishing the observed difference from the registered error-margin test: the measured contrast is $-3.286 \times 10^{-6}$ m, but its margin past the $1 \times 10^{-6}$ m threshold is $2.286 \times 10^{-6}$ m, below the $3.125 \times 10^{-6}$ m error floor. We therefore report the numerical difference without treating it as a resolved performance result. Additional weight conditions and coherent/incoherent diagnostics provide context; they are computational checks, not independent physical experiments.

## 2 Related Work

Finite-energy Airy beams and their propagation have been studied in the contexts of acceleration and beam control [1, 3]. Pearcey and butterfly fields are part of the broader class of catastrophe-related optical fields, with prior work addressing their propagation and focusing properties [2, 4].

Recent symmetric-butterfly studies concern tunable focusing and beam trajectories [7, 5]. The supplied evidence verifies metadata and publisher abstracts for these 2025 studies, but not their full texts. It is therefore insufficient for detailed comparison of field definitions, propagation conditions, observables, or results, and does not establish a novelty gap.

A study of partially coherent self-shifting cosine-Gaussian beams uses a cross-spectral-density model and considers propagation settings that include nonlinear media and radiation forces [6]. The incoherent arm examined here is a two-source intensity mixture in linear scalar free space, not a general partial-coherence model; it does not address nonlinear propagation or optical forces. This comparison clarifies scope rather than establishing novelty.

Rotational superposition and parity provide the closest conceptual context: inversion changes field symmetry, and odd moments can cancel under suitable conditions. The observable here is finite-interval centroid excursion for one apodized butterfly source, rather than core-energy concentration. This difference in observable and condition is descriptive and does not establish an independent contribution relative to existing parity or rotational-superposition work. Standard interference identities do not establish such an independent contribution either.

# 3 Theoretical Model

## 3.1 Apodized butterfly input and normalization

The source is the $n = 6$ butterfly slice. In dimensionless transverse coordinates $u = x/s$ and $v = y/s$, its canonical integral and apodized field are

$$C_n(u, v) = \int_{-\infty}^{\infty} \exp\left[i\left(t^n + vt^2 + ut\right)\right] dt, \qquad n = 6, \tag{1}$$

$$V(x, y) = C_6(x/s, y/s) \exp(-r^2/w^2), \qquad r^2 = x^2 + y^2. \tag{2}$$

The source scale is $s = 0.30$ mm. The Gaussian envelope radius $w = 0.30$ mm is the Gaussian envelope radius in the executed setup; unused Butterfly controls are zero. The field is zero outside the physical aperture of radius 0.9 mm.

For grid coordinates $x_j = (j - \lfloor N/2 \rfloor)L/N$ and $y_k = (k - \lfloor N/2 \rfloor)L/N$, the sampled input is normalized to total power $P$:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k). \tag{3}$$

Here $L$ is the grid extent, $N$ the number of samples per side, and $\Delta x = L/N$. The canonical integral is evaluated after contour rotation $t = \tau \exp[i\pi/(2n)]$, with $\tau$ uniformly sampled on $[-3, 3]$ and trapezoidal quadrature using 192 nodes. Source convergence uses 384 nodes.

## 3.2 Free-space propagation and composition

The executed propagation is scalar free-space angular-spectrum propagation with no lens applied:

$$U_z = \mathcal{F}^{-1}\left[\mathcal{F}(U_0) B_z \exp\left(i2\pi z \sqrt{\lambda^{-2} - f_x^2 - f_y^2}\right)\right]. \tag{4}$$

Here $f_x, f_y$ are transverse spatial frequencies and $\lambda$ is the wavelength. The conservative radial bandwidth cutoff is

$$\nu_c = \min\left[\lambda^{-1}, \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}\right], \qquad B_z(f_x, f_y) = \begin{cases} 1, & \sqrt{f_x^2 + f_y^2} \le \nu_c, \\ 0, & \sqrt{f_x^2 + f_y^2} > \nu_c. \end{cases} \tag{5}$$

This conservative radial restriction is not the full Matsushima passband.

Let $\mathcal{R}_\theta U$ denote rotation about the physical grid origin. The normalized coherent pair is

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2 \, dA}}\,(U + ae^{i\delta}\mathcal{R}_\theta U). \tag{6}$$

The primary composition uses $\theta = \pi$, $\delta = 0$, and $a = 0.8$. Rotation samples the real and imaginary fields independently by bilinear interpolation, with constant zero boundaries around the physical grid origin, and the rotated arm is normalized to input power. An exactly zero coherent field cannot be normalized; none of the measured conditions produces such a field.

For the incoherent diagnostic, the two arm intensities are added and normalized to the same total input power. If $A$ and $B = \mathcal{R}_\pi A$ denote the arm fields before pair normalization, the intensity accounting is

$$I_{\text{coh}} = \frac{P}{P_{\text{pair}}}\left(|A|^2 + a^2|B|^2 + 2a\,\mathrm{Re}\left[e^{i\delta}A^*B\right]\right), \tag{7}$$

$$I_{\text{inc}} = \frac{|A|^2 + a^2|B|^2}{1 + a^2}, \tag{8}$$

where $P_{\text{pair}}$ is the power of the unnormalized coherent sum. The normalized coherent and incoherent intensities therefore differ through both the cross term and normalization.

### 3.3 Conditional parity identity

Let $\mathsf{P}$ be a Hermitian involution, $\mathsf{P}^2 = I$, acting on a normalized state $u$, and let $Q$ be a Hermitian parity-odd operator satisfying $\mathsf{P}Q\mathsf{P} = -Q$. Define

$$q = \langle u, Qu\rangle, \qquad s = \langle u, \mathsf{P}u\rangle, \qquad \langle u, Q\mathsf{P}u\rangle = ij,$$

with real $q, s, j$. For real $a$ and phase $\delta$, set $g = u + ae^{i\delta}\mathsf{P}u$. Expanding the numerator and norm gives

$$\langle g, Qg\rangle = q(1 - a^2) - 2aj\sin\delta, \tag{9}$$

$$\langle g, g\rangle = 1 + a^2 + 2as\cos\delta. \tag{10}$$

Thus, if the norm is positive and the operator moments exist,

$$\frac{\langle g, Qg\rangle}{\langle g, g\rangle} = \frac{q(1 - a^2) - 2aj\sin\delta}{1 + a^2 + 2as\cos\delta}. \tag{11}$$

At zero phase, the overlap term $2as$ remains in the denominator. The simpler factor $(1-a^2)/(1+a^2)$ applies only when $s = 0$, or under a different normalization.

For centroid components, $Q$ is a transverse position operator. Equation (11) yields a conditional normalized-centroid relation for an inversion pair if propagation commutes with inversion and the moment domain is parity symmetric. It does not, by itself, establish a relation for every finite-window excursion: boundary truncation, interpolation, and finite axial sampling can affect that statistic. The saved trajectories are used below to check the relation for this sampled setup.

## 4 Numerical Methods

The reference calculation uses $N = 384$, extent $L = 2.4$ mm, wavelength $\lambda = 633$ nm, input power $P = 1$ W, and setup waist 0.30 mm. The primary propagation uses 11 equally spaced planes from 0.02 to 0.06 m. The post-review weight series and trajectory checks use 41 planes over the same interval. No lens is applied; the physical source aperture radius is 0.9 mm.

At each plane, the centroid is computed over a square detector domain $\Omega_h = [-h, h) \times [-h, h)$, centered on the physical grid origin:

$$\mathbf{r}_{\mathrm{cen}}(z) = \frac{\displaystyle\int_{\Omega_h} \mathbf{r}\,|U_z(\mathbf{r})|^2\,dA}{\displaystyle\int_{\Omega_h} |U_z(\mathbf{r})|^2\,dA}, \qquad \mathbf{r} = (x, y). \tag{12}$$

For the primary reference, $h = 1.2$ mm, half the grid extent. The excursion and signed contrast are

$$E_{\mathrm{cen}} = \max_{z\in\mathcal{Z}} \|\mathbf{r}_{\mathrm{cen}}(z) - \mathbf{r}_{\mathrm{cen}}(z_{\min})\|_2\,, \qquad \mathcal{Z} = \{z_j\}_{j=0}^{M-1} \subset [0.02, 0.06]\ \mathrm{m}, \tag{13}$$

$$\Delta E = E_{\mathrm{cen}}^{(\mathrm{coherent})} - E_{\mathrm{cen}}^{(\mathrm{single})}. \tag{14}$$

The registered target is $\Delta E \le -1.0 \times 10^{-6}$ m, conditional on resolved single-arm motion. The confirmation error floor is $3.125 \times 10^{-6}$ m. This numerical criterion is not experimental uncertainty.

As an ancillary diagnostic, the fixed-origin circular-core fraction is

$$C(z) = P^{-1} \int_{x^2+y^2\le r_c^2} |U_z|^2\,dA, \qquad \overline{C} = \mathrm{median}_{z\in\mathcal{Z}}\, C(z), \tag{15}$$

where $r_c = 0.11$ mm. The integral is evaluated by summing pixel intensities whose centers satisfy $r \le r_c$, then multiplying by $\Delta x^2$. This core-energy quantity is not the registered primary observable. Centroid-aligned core diagnostics recenter the circle at the plane centroid and are not substituted for the fixed-origin definition.

For independent propagation, the Fresnel field is evaluated by separable real-space quadrature of $e^{ikz}(i\lambda z)^{-1} \int U_0(\mathbf{r}') \exp[ik|\mathbf{r} - \mathbf{r}'|^2/(2z)]\,d^2\mathbf{r}'$, rather than by the angular-spectrum FFT. The same sampled input, output grid and aperture are used; agreement is tested within the paraxial regime. The confirmation checks include mesh convergence, two window checks, axial sampling, independent-observable validation, reference physics, coupled stochastic realization, quantization, held-out wavelength, and noise probes. The post-review suite evaluates weights $0, 0.4, 0.8, 1$ using six numerical tiers: reference, mesh, axial, Fresnel, window, and second window. The 24 weight-series solver evaluations are computational checks, not independent experiments. The reported sensitivity envelope across tiers is $8.913 \times 10^{-13}$ m. It describes deterministic variation among these calculations and does not replace the unchanged $3.125 \times 10^{-6}$ m error floor.

# 5 Results and Physical Interpretation

## 5.1 Registered primary comparison

At the reference tier, the single source has $E_{\mathrm{cen}} = 3.708 \times 10^{-6}$ m and the coherent weight-0.8 pair has $E_{\mathrm{cen}} = 0.4213 \times 10^{-6}$ m. The contrast is

$$\Delta E = -3.2864 \times 10^{-6}\ \mathrm{m}.$$

This is a nonzero numerical difference whose magnitude exceeds the registered $1.0 \times 10^{-6}$ m threshold. The margin beyond the threshold, however, is $2.2864 \times 10^{-6}$ m, below the $3.125 \times 10^{-6}$ m error floor. The registered error-margin criterion is not passed. The numerical result is therefore a reproduced lower excursion value, not an established resolved reduction under the registered criterion.

Table 1 reports the paired contrast across the six registered post-review numerical tiers. The tiers agree closely, but are not independent observations and do not override the error-margin criterion.

Table 1: Post-review primary coherent-minus-single contrast at weight 0.8, computed directly from the saved paired measurements. $L$ is the square computational and integration side; $M$ is the number of axial planes. All rows use angular-spectrum propagation except the independently computed real-space Fresnel row. $\delta\Delta E$ is the difference from the first row. These are numerical diagnostics, not independent experimental replicates.

| Tier | $N$ | $L$ (mm) | $M$ | $\Delta E$ ($\mu$m) | $\delta\Delta E$ (pm) |
|---|---|---|---|---|---|
| Reference | 384 | 2.4 | 11 | -3.286437915 | 0.000 |
| Mesh | 768 | 2.4 | 11 | -3.286437914 | 0.001 |
| Axial | 768 | 2.4 | 41 | -3.286437914 | 0.001 |
| Fresnel | 768 | 2.4 | 41 | -3.286437023 | 0.891 |
| Window | 1152 | 3.6 | 41 | -3.286437808 | 0.106 |
| Second window | 1536 | 4.8 | 41 | -3.286437787 | 0.128 |

Table 2: Excursions from the 41-plane weight series. The source, wavelength, aperture, power, propagation interval, and centroid definition are matched across arms. Contrast means coherent minus single. The zero-weight case is an identity control; the equal-weight case is a symmetry-limit diagnostic.

| Weight $a$ | Single ($\mu$m) | Coherent ($\mu$m) | Incoherent ($\mu$m) | Contrast ($\mu$m) |
|---|---|---|---|---|
| 0.0 | 3.707744 | 3.707744 | 3.707744 | 0 |
| 0.4 | 3.707744 | 1.618675 | 2.684918 | $-2.089069$ |
| 0.8 | 3.707744 | 0.421306 | 0.813895 | $-3.286438$ |
| 1.0 | 3.707744 | $\simeq 0$ | $\simeq 0$ | $-3.707744$ |

### 5.2 Weight series and coherence diagnostic

The post-review 41-plane weight-series excursions are given in Table 2. At $a = 0$, the composed arm coincides with the single source, as expected for the identity control. At $a = 1$, the coherent inversion pair has essentially zero excursion, consistent with the inversion-even limit. The intermediate weights lie between these endpoints in the measured series. These four conditions describe this setup; they do not establish a general weight law.

At $a = 0.8$, the incoherent arm also has a much smaller excursion than the single source. Its excursion is $0.814 \times 10^{-6}$ m, compared with $0.421 \times 10^{-6}$ m for the coherent pair. The coherent-minus-incoherent difference is $-0.393 \times 10^{-6}$ m, smaller than the $3.125 \times 10^{-6}$ m error floor. These diagnostics do not show that coherence is necessary for reduced excursion or resolve a coherence-specific advantage.

### 5.3 Conditional relation and detector-window sensitivity

Under the assumptions of the parity calculation, the zero-phase coherent centroid scale factor is

$$k_{\mathrm{coh}} = \frac{1 - a^2}{1 + a^2 + 2a\,\mathrm{Re}(s)},$$

where $s = \langle u, \mathsf{P}u \rangle$ is the normalized input overlap. For the incoherent mixture, the corresponding factor is

$$k_{\mathrm{inc}} = \frac{1 - a^2}{1 + a^2}.$$

For the weight-0.8 input, the measured real overlap is 0.955135, giving $k_{\mathrm{coh}} = 0.113629$ and $k_{\mathrm{inc}} = 0.219512$. Using these factors with the single-source trajectory predicts the measured coherent and incoherent trajectories without fitting to them. Across the saved trajectory

comparisons, the maximum displacement discrepancy is $6.01 \times 10^{-15}$ m. Transfer of a centroid scale factor to the excursion requires it to remain constant across the sampled planes: if $\mathbf{c}_a(z_j) = k_a \mathbf{c}_0(z_j)$ for all $j$, then $E_a = |k_a| E_0$. Constancy requires conservation of the relevant power and overlap; a finite detector or the bandwidth mask can violate it. The saved-trajectory discrepancy tests this conditional reduction for the executed finite domains without changing the original error gate. This agreement is a source-specific numerical check of a conditional identity, not evidence for a new parity law or a general weight response. Its interpretation remains conditional on the stated symmetry, propagation, moment-domain, and normalization assumptions.

The primary contrast was also recalculated for detector half-widths $h = 1.2,\ 1.8,\ 2.4$ mm, using the half-open square domains $[-h, h) \times [-h, h)$. All three detector domains are evaluated on the same propagated $N = 1536$, $L = 4.8$ mm field with $M = 41$ planes, so only the integration mask changes. This detector check is separate from enlarging the propagation grid. The respective contrasts are $-3.286437836$, $-3.286437791$, and $-3.286437787 \times 10^{-6}$ m. This shows stability over the tested square windows, not over arbitrary domains. The small spread across these detector windows and the tier sensitivity envelope are numerical sensitivities, not experimental or statistical uncertainty estimates.

Figure 1 shows the weight conditions, matched centroid trajectories, all numerical contrasts and the separate detector-domain test. The line segments joining sampled weights guide the eye; they are not additional measurements or a fitted law.

### 5.4 Interpretation and scope

Incomplete cancellation of odd moments is a plausible account of the unequal-weight response. The normalized coherent expectation includes the overlap in its denominator, and the conditional centroid relation matches the saved trajectories for the tested cases. Yet the primary statistic is a maximum displacement over a finite axial interval and detector domain, not an abstract full-plane operator expectation. The agreement does not establish the assumptions for other sampled fields or turn the identity into a general excursion law.

The incoherent arm also reduces excursion substantially, so the reduction relative to the single source cannot be assigned uniquely to coherent interference. At weight 0.8, the coherent/incoherent difference is about $0.393 \times 10^{-6}$ m, below the reported error floor. The normalized intensity relation in Eq. (8) accounts for cross-term and global-normalization effects; those identities are not independent evidence for a separate mechanism.

The closest symmetric-butterfly references are available here through verified metadata and, for Lin et al., a publisher abstract; a full-text comparison remains unavailable [7, 5]. Accordingly, no novelty gap is claimed. The results apply to a numerical scalar free-space model with the stated source, aperture, wavelength, and observation interval. No laboratory measurements are reported, and numerical tiers do not quantify experimental uncertainty.

## 6 Conclusions

For the executed apodized butterfly source, a coherent 180° inversion pair with zero phase and weight 0.8 has a reference centroid excursion of $0.421 \times 10^{-6}$ m, compared with $3.708 \times 10^{-6}$ m for the single source. The contrast, $-3.286 \times 10^{-6}$ m, is reproduced across the reported numerical checks and is stable over the tested detector windows. It does not pass the registered error-margin criterion: the $2.286 \times 10^{-6}$ m margin beyond the target is smaller than the $3.125 \times 10^{-6}$ m error floor.

The weight series and conditional centroid relation provide setup-specific context, but do not establish a general weight law. The incoherent mixture also reduces excursion, so the evidence does not show that coherence is necessary. The conditional parity identity, with overlap-dependent

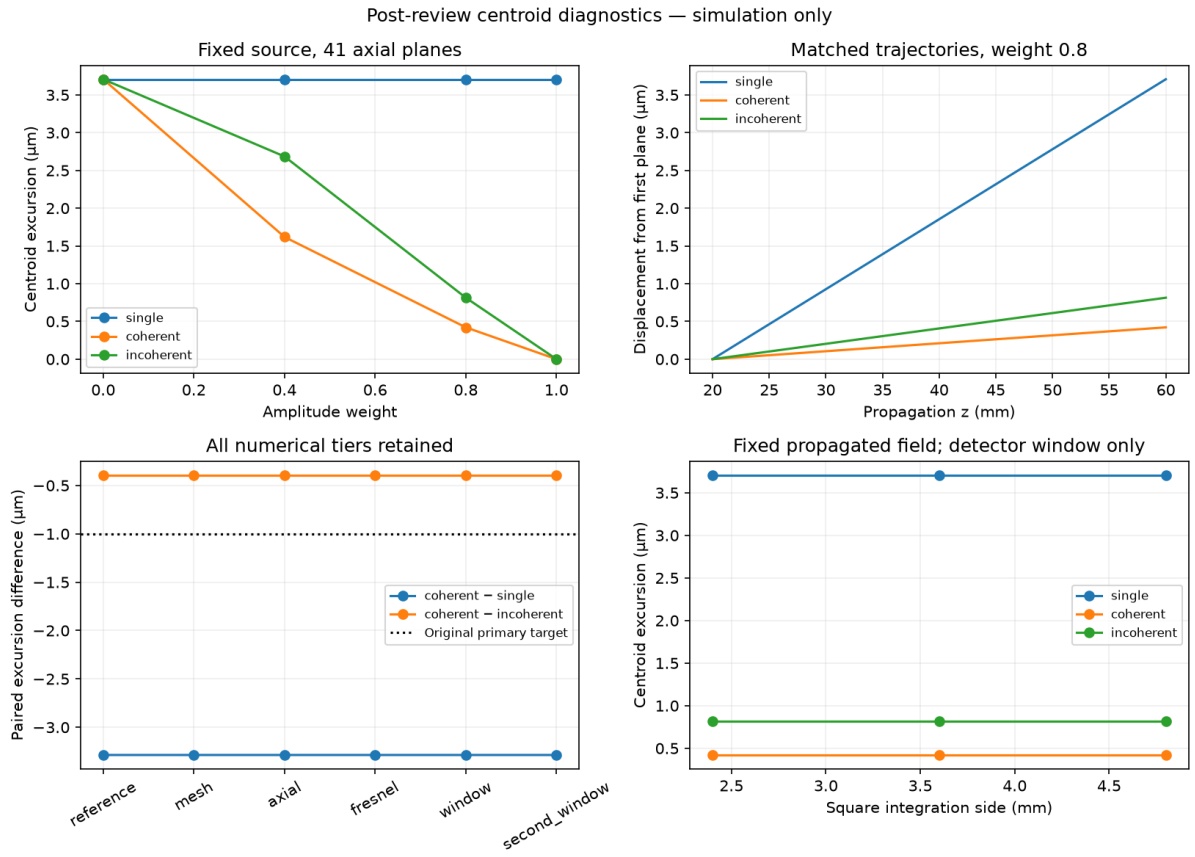



Figure 1: Centroid diagnostics at 633 nm, 1 W and a 0.9 mm input-aperture radius. Top left: single, coherent and incoherent excursion at weights 0, 0.4, 0.8 and 1, using $N = 768$, $L = 2.4$ mm and 41 axial planes. Top right: displacement from each arm's first-plane centroid for weight 0.8 over 20–60 mm. Bottom left: coherent-minus-single and coherent-minus-incoherent contrasts over all six registered numerical tiers; the dotted line is the original $-1\,\mu$m primary target, and does not include the separate $3.125\,\mu$m error requirement. Bottom right: excursions over detector squares of side 2.4, 3.6 and 4.8 mm, all integrated from the same $N = 1536$, $L = 4.8$ mm propagated field. Apparent overlap of tier results is quantified in Table 1.

normalization, agrees with the saved trajectories under its assumptions but does not establish a general mechanism. These results are numerical observations for the stated configuration; they establish neither practical superiority nor novelty relative to the closest symmetric-butterfly literature.

## Disclosures

The numerical study and manuscript preparation used the Auto BeamScientist system. No experimental measurements are reported.

## References

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